# PH 724 Data Analysis & Scientific Communication Incorporating Use of Generative AI

Part 10: Binomial GLMs, Binomial GLMMs, Quasi-binomial and Beta-binomial Models

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## Motivating example: bumblebee infection

Define aggregated responses

$$S_i = \sum_{j=1}^{n_i} Y_{ij}$$

- $i$ indexes colonies
- $j$ indexes bees within colony $i$
- $Y_{ij}$ is a binary indicator (1 = infected, 0 = not infected)
- $n_i$ is the number of observed bees in colony $i$ at the end of experiment

## 1. Binomial generalized linear model (GLM)

$$S_i \sim \text{Binomial}(n_i, p_i),$$
$$g(p_i) = \boldsymbol{x}_i^\top \boldsymbol{\beta}.$$

- $S_i$: number of infected bees in colony $i$
- $n_i$: total observed bees in that colony
- $p_i$: probability of infection for a bee in colony $i$
- Binomial$(n_i, p_i)$: binomial distribution with $n_i$ trials and success probability $p_i$.
- $g$: a link function
- $\boldsymbol{x}_i$: the vector of covariates (including the intercept and various indicators).
- $\boldsymbol{\beta}$: the vector of fixed effects.

Accordingly,

$$p_i = g^{-1}(\boldsymbol{x}_i^\top \boldsymbol{\beta}), \qquad E(S_i) = n_i p_i, \qquad \text{Var}(S_i) = n_i p_i (1 - p_i).$$

With the binomial specification, individual bee outcomes are independent conditional $p_i$. This can be implausible for bees sharing a colony. An aggregated binomial likelihood does not automatically account for extra variation due to within-colony dependence.

### Available link functions for binomial GLMs

The link function $g$ connecting the probability $p_i$ to the linear predictor $\boldsymbol{x}_i^\top \boldsymbol{\beta}$. Changing the link changes how covariates relate to $p_i$, but does not by itself account for overdispersion or within-colony dependence. Logistic regression is the binomial GLM with the logit link; other links are possible.

| Link | Mathematical form $g(p)$ | R specification |
|---|---|---|
| Logit (default) | $\log\{p/(1-p)\}$ | `binomial(link = "logit")` |
| Probit | $\Phi^{-1}(p)$ | `binomial(link = "probit")` |
| Log | $\log(p)$ | `binomial(link = "log")` |
| Complementary log-log | $\log\{-\log(1-p)\}$ | `binomial(link = "cloglog")` |
| Cauchit | $\tan\{\pi(p-1/2)\}$ | `binomial(link = "cauchit")` |

## 2. Binomial generalized linear mixed-effects model (GLMM)

A GLMM adds random effects to GLM to allow dependence across colonies:

$$S_i \mid \boldsymbol{\gamma}_i \sim \text{Binomial}(n_i, p_i),$$
$$g(p_i) = \boldsymbol{x}_i^\top \boldsymbol{\beta} + \boldsymbol{z}_i^\top \boldsymbol{\gamma}_i, \qquad \boldsymbol{\gamma}_i \overset{\text{iid}}{\sim} \text{MVN}(0, \Sigma_\gamma).$$

- $\boldsymbol{z}_i$: the vector of covariates corresponding to random effects.
- $\boldsymbol{\gamma}$: the vector of random effects.

## 3. Quasi-binomial model

A quasi-binomial model adjusting variance without specifying a new model for $p_i$. Specifically, the quasi-binomial GLM is

$$g(p_i) = \boldsymbol{x}_i^\top \boldsymbol{\beta}, \qquad E(S_i) = n_i p_i, \qquad \text{Var}(S_i) = \phi n_i p_i (1-p_i);$$

The quasi-binomial GLMM is

$$g(p_i) = \boldsymbol{x}_i^\top \boldsymbol{\beta} + \boldsymbol{z}_i^\top \boldsymbol{\gamma}_i, \qquad E(S_i \mid \boldsymbol{\gamma}_i) = n_i p_i, \qquad \text{Var}(S_i \mid \boldsymbol{\gamma}_i) = \phi n_i p_i (1-p_i), \qquad \boldsymbol{\gamma}_i \overset{\text{iid}}{\sim} \text{MVN}(0, \Sigma_\gamma).$$

- $\phi$ is an estimated dispersion factor. For $\phi$ greater than 1, quasi-binomial standard errors are typically larger than ordinary binomial standard errors.
- The quasi-binomial GLM may be fitted via `R` function `glm` with `family = quasibinomial(link = "logit")`.
- The quasi-binomial GLMM may be fitted via `R` function `MASS::glmmPQL` with `family = quasibinomial(link = "logit")`.

## 4. Beta-binomial model

$$S_i \mid P_i \sim \text{Binomial}(n_i, P_i),$$
$$P_i \sim \text{Beta}\left(\mu_i \frac{1-\rho}{\rho}, (1-\mu_i)\frac{1-\rho}{\rho}\right),$$
$$\text{logit}(\mu_i) = \boldsymbol{x}_i^\top \boldsymbol{\beta}.$$

Or, equivalently,

$$S_i \sim \text{BetaBinomial}(n_i, \mu_i, \rho),$$
$$\text{logit}(\mu_i) = \boldsymbol{x}_i^\top \boldsymbol{\beta}.$$

Accordingly,

$$E(S_i) = n_i \mu_i,$$
$$\text{Var}(S_i) = n_i \mu_i (1-\mu_i)\{1 + (n_i - 1)\rho\}.$$

- $P_i$: the random infection probability for colony $i$
- $\mu_i$: mean infection probability for colony $i$.

- $\rho \in (0, 1)$: the within-colony correlation. Unlike a constant quasi-binomial dispersion, the beta-binomial variance inflation depends on colony size $n_i$.
- Model fitting may be implemented via `R` function `glmmTMB::glmmTMB` with `family = betabinomial(link = "logit")`.